

M – Tech Production & Marketing

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REPORT OF EXPERIMENT 11

Natural Language Processing — NLP, Transformers & BERT

Introduction

This report summarizes the activities completed during **Experiment 11** of the M-Tech Production & Marketing AI/ML Internship Program. The experiment focused on the fundamentals of **Natural Language Processing (NLP)**. A Python program was developed to process text by converting it to lowercase, splitting it into words, counting words and characters, and identifying unique words. This practical activity introduced the basic concepts of text processing used in AI applications.

Objectives

- To understand the basics of Natural Language Processing.
- To process textual data using Python.
- To convert text into lowercase.
- To split sentences into words.
- To count words and characters.
- To identify unique words in a sentence.

Task Description

The assigned task was to develop a Python application that performs basic Natural Language Processing operations. The program accepts a sentence from the user, converts it to lowercase, separates the text into individual words, counts the total

number of words and characters, identifies unique words, and displays the processed information.

Work Done

Today, I developed a simple NLP application using Python. I created a program that accepts user input, performs text preprocessing by converting the sentence into lowercase, separates the words, calculates the total number of words and characters, and identifies unique words. After testing the application with different sentences, all outputs were generated successfully.

Challenges Faced

During the experiment, I encountered minor issues related to text formatting and counting unique words. These issues were resolved by using Python string methods and the set() function. After testing the program with different inputs, all functions worked correctly.

Key Learnings

This experiment enhanced my understanding of Natural Language Processing and text preprocessing techniques. I learned how to process text, convert it into lowercase, split sentences into words, count words and characters, and identify unique words. These concepts form the foundation of many AI applications such as chatbots, sentiment analysis, and language translation.

Conclusion

The experiment was completed successfully, and all objectives were achieved. The practical implementation strengthened my understanding of basic NLP techniques and provided hands-on experience in processing textual data using Python. This experiment serves as a foundation for advanced NLP models such as Transformers and BERT.

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